

FORMULARIO RETI LOGICHE COMBINATORIE

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TEOREMI DELL'ALGEBRA BOOLEANA

NOME	FORMA AND	FORMA OR
IDENTITÀ	$1A = A$	$0 + A = A$
NULLABILITÀ	$0A = 0$	$1 + A = 1$
IDEMPOTENZA	$AA = A$	$A + A = A$
COMPLEMENTARITÀ	$A\bar{A} = 0$	$A + \bar{A} = 1$
COMMUTATIVA	$AB = BA$	$A + B = B + A$
ASSOCIATIVA	$(AB)C = A(BC)$	$(A + B) + C = A + (B + C)$
DISTRIBUTIVA	$A + BC = (A + B)(A + C)$	$A(B + C) = AB + AC$
ASSORBIMENTO	$A(A + B) = A$	$A + AB = A$
DE MORGAN	$\overline{AB} = \bar{A} + \bar{B}$	$\overline{A + B} = \bar{A}\bar{B}$

PORTE LOGICHE E ESPRESSIONI BOOLEANE

DISEGNO	PORTA LOGICA	SIMBOLO	ESPRESSIONE	FORMA ALT.
	AND	$A \cdot B$	AB	
	OR	$A + B$	$A + B$	
	NOT	$\bar{A}$	$\bar{A}$	
	NAND	$\overline{AB}$	$\overline{AB}$	$\bar{A} + \bar{B}$
	NOR	$\overline{A + B}$	$\overline{A + B}$	$\bar{A}\bar{B}$
	XOR	$A \oplus B$	$\bar{A}B + A\bar{B}$	
	XNOR	$A \odot B$	$AB + \bar{A}\bar{B}$	
	BUFFER	$-$	A	VALORE INALTERATO

NOZIONI DA RICORDARE

- SE DUE O PIÙ PROBLEMI HANNO LO STESSO NUMERO DI VARIABILI IN ENTRATA, SI CREA UN'UNICA TABELLA.
- NON TUTTI I PROCEDIMENTI DI APPLICAZIONE DEI TEOREMI PORTANO ALLA FORMA MINIMA.
IN QUESTO CASO BISOGNA:
1) DERIVARE LA FORMA MINIMA DALLA TABELLA DI VERITÀ O K-MAP.
2) OPPURE CERCARE FIN DALL'INIZIO DI OTTENERE TERMINI DELLA STESSA LUNGHEZZA.
- PER POPOLARE LA TABELLA DI VERITÀ L'ESPRESSIONE DEVE TROVARSI IN FORMA POS OPPURE SOP.
- PER QUANTO RIGUARDA LA COSTRUZIONE DELLA TABELLA DI VERITÀ E SUCCESSIVA DERIVAZIONE DELLA FORMA MINIMA, BISOGNA STARE ATTENTI ALLA FORMA CON CUI SI STA LAVORANDO:

IN GENERALE:

• SOP (SOMME DI PRODOTTI)

- $A = 1$
- $\bar{A} = 0$
- $S = 1$

ESEMPIO:

A	B	C	D	S
0	0	1	1	1
0	1	0	1	1

$$\rightarrow \bar{A}\bar{B}CD + \bar{A}B\bar{C}D$$

• POS (PRODOTTI DI SOMME)

- $A = 0$
- $\bar{A} = 1$
- $S = 0$

ESEMPIO:

A	B	C	D	S
0	0	1	1	0
0	1	0	1	0

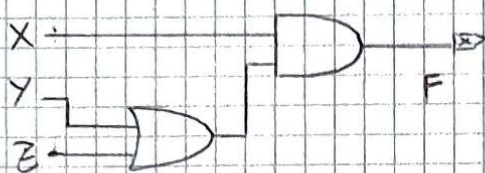
$$\rightarrow (A+B+\bar{C}+\bar{D}) \times (A+\bar{B}+C+\bar{D})$$

RETI LOGICHE COMBINATORIE

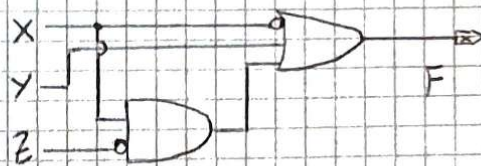
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1) ESERCIZIO 1 - SLIDES

A) $F = X(Y + Z)$



B) $F = \bar{X} + Y + X\bar{Z}$



2) ESERCIZIO 2 - SLIDES

A) RICAVO LA FORMULA

$$Z = Z_1 + Z_2$$

$$Z_1 = \bar{Z}_3 + \bar{Z}_4$$

$$Z_2 = \bar{C} + \bar{D}$$

$$Z_3 = A + B$$

$$Z_4 = B + C + \bar{D}$$

$$= (A+B) + (B+C+\bar{D}) + \bar{C} + \bar{D}$$

$$= (A+B) + (B+C+\bar{D})$$

SEMPLIFICHIAMO L'ESPRESSIONE

$$Z = \bar{A} \cdot \bar{B} + \bar{B} \bar{C} \bar{D} + \bar{C} + \bar{D}$$

$$= \bar{A} \cdot \bar{B} + \bar{C} (1 + \bar{B} \bar{D}) + \bar{D}$$

$$= \bar{A} \cdot \bar{B} + \bar{C} + \bar{D}$$

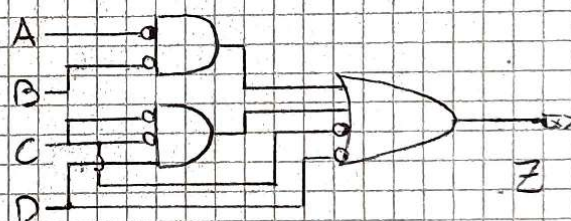
ASSORBIMENTO
INVERSO

$$\bar{C} = \bar{C} + \bar{C} \bar{B} \bar{D}$$

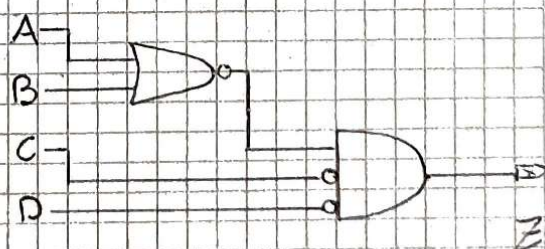
TABELLA DI VERITÀ

A	B	C	D	Z
0	0	0	0	1
0	0	0	1	1
0	0	1	0	1
0	0	1	1	1
0	1	0	0	1
0	1	0	1	1
0	1	1	0	1
0	1	1	1	0
1	0	0	0	1
1	0	0	1	1
1	0	1	0	1
1	0	1	1	0
1	1	0	0	1
1	1	0	1	1
1	1	1	0	1
1	1	1	1	0

RETE RISULTANTE TOTALE



RETE RISULTANTE MINIMA



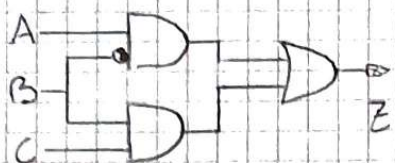
$$(\bar{A} + \bar{C} + \bar{D})(B + C + \bar{D})$$

FORMA POS

$$B) Z = Z_1 + Z_2 = (A\bar{B}) + (BC)$$

$$Z_1 = A\bar{B}$$

$$Z_2 = BC$$



A	B	C	Z	A	B	C	Z
0	0	0	0	1	0	0	1
0	0	1	0	1	0	1	1
0	1	0	0	1	1	0	0
0	1	1	1	1	1	1	1

$A\bar{B}$

BC

3) ESERCIZIO 3 - SLIDES

$$A) Z_A = Z_1 + Z_2 + Z_3 = ABC + A\bar{B}\bar{C} + A\bar{B}C$$

$$Z_1 = ABC$$

$$= AB(\bar{C} + C) + A\bar{B}C$$

$$Z_2 = A\bar{B}\bar{C}$$

$$= A(\bar{B} + \bar{B}C)$$

$$Z_3 = A\bar{B}C$$

$$= A(\bar{B} + BC + \bar{B}C)$$

$$= A(\bar{B} + C(\bar{B} + B))$$

$$= AB + AC$$

DISTRIBUZIONE

COMPLEMENTARIETÀ $C + \bar{C} = 1$

DISTRIBUZIONE

ASSORBIMENTO $B = B + BC$

COMPLEMENTARIETÀ $B + \bar{B} = 1$

$$B) Z_B = A(B + C) = AB + AC$$

4) ESERCIZIO 4 - SLIDES

$$A) Z_1 = \bar{A}\bar{B} + \bar{C}\bar{D} + \bar{B}D + C$$

$$= \bar{A}\bar{B} + \bar{B}D + C + \bar{D}(\bar{C} + C)$$

$$= \bar{A}\bar{B} + C + \bar{D} + \bar{B}\bar{D} + \bar{B}D$$

$$= \bar{A}\bar{B} + C + \bar{D} + \bar{B}(\bar{D} + D)$$

$$= \bar{A}\bar{B} + \bar{B} + C + \bar{D}$$

$$= \bar{A}\bar{B} + \bar{B} + \bar{A}\bar{B} + C + \bar{D}$$

$$= \bar{B} + \bar{A}(\bar{B} + B) + C + \bar{D} = \bar{A} + \bar{B} + C + \bar{D}$$

ASSORBIMENTO $C = C + C\bar{D}$

COMPLEMENTARIETÀ

ASSORBIMENTO $\bar{B} = \bar{B} + \bar{B}B$

COMPLEMENTARIETÀ

ASSORBIMENTO $\bar{B} = \bar{B} + \bar{B}A$

A	B	C	D	Z ₁	A	B	C	D	Z ₁	A	B	C	D	Z ₁	A	B	C	D	Z ₁
0	0	0	0	1	0	1	0	0	1	1	0	0	0	1	1	1	0	0	1
0	0	0	1	1	0	1	0	1	1	1	0	0	1	1	1	1	0	1	0
0	0	1	0	1	0	1	1	0	1	1	0	1	0	1	1	1	1	0	1
0	0	1	1	1	0	1	1	1	1	1	0	1	1	1	1	1	1	1	1

$$\bar{A} + \bar{B} + C + \bar{D}$$

$$B) Z_2 = A + \bar{B}(\bar{C} + D)$$

DISTRIBUZIONE

$$= A + \bar{B}\bar{C} + \bar{B}D$$

A	B	C	D	Z ₂	A	B	C	D	Z ₂	A	B	C	D	Z ₂	A	B	C	D	Z ₂
0	0	0	0	1	0	1	0	0	0	1	0	0	0	1	1	1	0	0	1
0	0	0	1	1	0	1	0	1	0	1	0	0	1	1	1	1	0	1	1
0	0	1	0	0	0	1	1	0	0	1	0	1	0	1	1	1	1	0	1
0	0	1	1	1	0	1	1	1	0	1	0	1	1	1	1	1	1	1	1

RETI LOGICHE COMBINATORIE

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$$C) Z_3 = AB + C(\bar{D} + B) + D(\bar{B} + \bar{C})$$

DISTRIBUZIONE

$$= AB + C\bar{D} + CB + D\bar{B} + D\bar{C}$$

A	B	C	D	Z ₃	A	B	C	D	Z ₃	A	B	C	D	Z ₃	A	B	C	D	Z ₃
0	0	0	0	0	0	1	0	0	0	1	0	0	0	0	1	1	0	0	1
0	0	0	1	1	0	1	0	1	1	1	0	0	1	1	1	1	0	1	1
0	0	1	0	1	0	1	1	0	1	1	0	1	0	1	1	1	1	0	1
0	0	1	1	1	0	1	1	1	1	1	0	1	1	1	1	1	1	1	1

$$D) Z_4 = ABC + \bar{C}(\bar{A} + \bar{B} + C)$$

$$= ABC + \bar{A}\bar{C} + \bar{B}\bar{C} + \bar{C}C$$

$$= \boxed{ABC + \bar{A}\bar{C} + \bar{B}\bar{C}}$$

A	B	C	Z ₄	A	B	C	Z ₄
0	0	0	1	1	0	0	1
0	0	1	0	1	0	1	0
0	1	0	1	1	1	0	0
0	1	1	0	1	1	1	1

ESERCIZIO 5 - SLIDES

$$Y_1 = (A+B)(A+BC) + \bar{A}\bar{B} + \bar{A}\bar{C}$$

DISTRIBUZIONE

$$\cancel{A}A + ABC + BA + BC + \bar{A}\bar{B} + \bar{A}\bar{C}$$

$$A + ABC + BA + BC + \bar{A}\bar{B} + \bar{A}\bar{C}$$

$$A(1 + BC + B) + BC + \bar{A}\bar{B} + \bar{A}\bar{C}$$

$$A + \bar{A}\bar{B} + \bar{A}\bar{C} + BC + \bar{A}\bar{B} + \bar{A}\bar{C}$$

$$A + \bar{B}(\cancel{A+A}) + \bar{C}(\cancel{A+A}) + BC$$

$$A + \bar{B} + \bar{C} + BC$$

$$A + \bar{B} + \bar{C} + \bar{C}B + BC$$

$$A + \bar{B} + \bar{C} + B(\cancel{C+C})$$

$$A + \bar{B} + \bar{C} + B$$

$$A + 1 + \bar{C} = \boxed{1}$$

IDEMPOTENZA $A \cdot A = A$

ASSORBIMENTO INVERSO

ASSORBIMENTO $A = A + A\bar{B} + A\bar{C}$

COMPLEMENTARIETÀ $A \cdot \bar{A} = 0$

ASSORBIMENTO $\bar{C} = \bar{C} + \bar{C}B$

COMPLEMENTARIETÀ $C + \bar{C} = 1$

COMPLEMENTARIETÀ $\bar{B} + B = 1$

ELEMENTI NULLI

$$Y_2 = ABC + AB\bar{C} + \bar{A}BD + ABD + \bar{A}D$$

$$AB(\cancel{C+C}) + \bar{A}BD + ABD + \bar{A}D$$

$$AB(\cancel{1+D}) + \bar{A}D(\cancel{1+B})$$

$$\boxed{AB + \bar{A}D}$$

COMPLEMENTARIETÀ

ASSORBIMENTO INVERSO

$$Y_3 = (X + Y + \bar{W})(X + Y + W)(X + \bar{Y} + W)(\bar{X} + \bar{Y} + W)$$

$$= \boxed{(X + Y)(\bar{Y} + W)}$$

COMBINAZIONE

$$\begin{aligned}
 Y_4 &= \bar{A}C(\bar{A}B\bar{D}) + \bar{A}B\bar{C}\bar{D} + A\bar{B}C \\
 &\bar{A}C(A + \bar{B} + \bar{D}) + \bar{A}B\bar{C}\bar{D} + A\bar{B}C \\
 &\bar{A}\bar{A}C + \bar{A}\bar{B}C + \bar{A}\bar{D}C + \bar{A}B\bar{C}\bar{D} + A\bar{B}C \\
 &\bar{A}\bar{B}C + \bar{A}C\bar{D} + \bar{A}B\bar{C}\bar{D} + A\bar{B}C \\
 &\bar{B}C(\bar{A} + \bar{A}) + \bar{A}C\bar{D} + \bar{A}B\bar{C}\bar{D} \\
 &\bar{B}C + \bar{A}\bar{D}(C + \bar{C}B) \\
 &\bar{B}C + \bar{A}\bar{D}(C + CB + \bar{C}B) \\
 &\bar{B}C + \bar{A}\bar{D}(C + B(\bar{A} + \bar{C})) \\
 &\bar{B}C + \bar{A}\bar{D}C + \bar{A}\bar{D}B
 \end{aligned}$$

NO FORMA MINIMA

DOBBIAMO OTTENERE IMPLICANTI CON EGUAL NUMERO DI TERMINI:

$$\begin{aligned}
 &\bar{A}\bar{B}C + \bar{A}C\bar{D} + \bar{A}B\bar{C}\bar{D} + A\bar{B}C \\
 &\bar{A}\bar{B}C(D + \bar{D}) + \bar{A}C\bar{D}(B + \bar{B}) + \bar{A}B\bar{C}\bar{D} + A\bar{B}C(D + \bar{D}) \\
 &\bar{A}\bar{B}CD + \bar{A}\bar{B}C\bar{D} + \bar{A}C\bar{D}B + \bar{A}C\bar{D}\bar{B} + \bar{A}B\bar{C}\bar{D} \\
 &\quad + A\bar{B}CD + A\bar{B}C\bar{D} \\
 &\bar{B}C(\bar{A}D + \bar{A}\bar{D} + AD + A\bar{D}) + \bar{A}B\bar{C}\bar{D} + \bar{A}B\bar{C}\bar{D} \\
 &\bar{B}C(\bar{A}(D + \bar{D}) + A(D + \bar{D})) + \bar{A}B\bar{C}\bar{D} + \bar{A}B\bar{C}\bar{D} \\
 &\bar{B}C(\bar{A} + A) + \bar{A}B\bar{C}\bar{D} + \bar{A}B\bar{C}\bar{D} \\
 &\bar{B}C + \bar{A}B\bar{C}\bar{D} + \bar{A}B\bar{C}\bar{D} \\
 &\bar{B}C + \bar{A}B\bar{D}(C + \bar{C}) \\
 &\bar{B}C + \bar{A}B\bar{D}
 \end{aligned}$$

A	B	C	D	Y_4
0	0	0	0	0
0	0	0	1	0
0	0	1	0	1
0	0	1	1	1
0	1	0	0	1
0	1	0	1	0
0	1	1	0	1
0	1	1	1	0
1	0	0	0	0
1	0	0	1	0
1	0	1	0	1
1	0	1	1	1
1	1	0	0	0
1	1	0	1	0
1	1	1	0	0
1	1	1	1	0

$$\bar{A}B\bar{D} + \bar{B}C$$

DE MORGAN

DISTRIBUZIONE

COMPLEMENTARIETÀ $A \cdot \bar{A} = 0$

DISTRIBUZIONE

COMPLEMENTARIETÀ $A + \bar{A} = 1$

DISTRIBUZIONE

ASSORBIMENTO $C = C + CB$

COMPLEMENTARIETÀ $C + \bar{C} = 1$

COMPLEMENTARIETÀ & IDEMPOTENZA

COMPLEMENTARIETÀ

ELEMENTI NULLI

$$\begin{aligned}
 Y_5 &= (\bar{A} + B)(A + B + D)\bar{D} \\
 &= \bar{D}(\bar{A}A + \bar{A}B + \bar{A}D + AB + B\bar{D} + BD) \\
 &= \bar{A}B\bar{D} + \bar{A}\bar{D}D + AB\bar{D} + B\bar{D} + B\bar{D}D \\
 &= \bar{A}B\bar{D} + AB\bar{D} + B\bar{D} \\
 &= B\bar{D}(\bar{A} + A + 1) \\
 &= B\bar{D}
 \end{aligned}$$

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$$Y_6 = \bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}C\bar{D} + \bar{A}B\bar{C}D + A\bar{B}\bar{C}D$$

$$\bar{A}\bar{B}D(\bar{C} + C) + \bar{A}\bar{B}C\bar{D} + A\bar{B}\bar{C}D$$

$$\bar{A}\bar{B}D(\bar{B} + B) + \bar{A}\bar{B}C\bar{D} + A\bar{B}\bar{C}D$$

$$\bar{A}\bar{B}D(\bar{B} + \bar{C}(\bar{B} + B)) + \bar{A}\bar{B}C\bar{D} + A\bar{B}\bar{C}D$$

$$\bar{A}\bar{B}D + \bar{A}\bar{C}D + \bar{A}B\bar{C}D + A\bar{B}\bar{C}D$$

$$\bar{B}D(\bar{A} + A\bar{C}) + \bar{A}\bar{C}D + A\bar{B}\bar{C}D$$

$$\bar{B}D(\bar{A} + A\bar{C} + A\bar{C}) + \bar{C}D(\bar{A} + A\bar{B})$$

$$\bar{B}D(\bar{A} + \bar{C}) + \bar{C}D(\bar{A} + A\bar{B} + A\bar{B})$$

$$\bar{A}\bar{B}D + \bar{B}\bar{C}D + \bar{C}D(\bar{A} + \bar{B}(\bar{A} + A))$$

$$\bar{A}\bar{B}D + \bar{B}\bar{C}D + \bar{B}\bar{C}D + \bar{A}\bar{C}D$$

$$\bar{A}\bar{B}D + \bar{B}\bar{C}D + \bar{A}\bar{C}D$$

COMPLEMENTARIETÀ $C + \bar{C} = 1$

ASSORBIMENTO $\bar{B} = \bar{B} + \bar{B}C$

COMPLEMENTARIETÀ $B + \bar{B} = 1$

DISTRIBUZIONE

ASSORBIMENTO $\bar{A} = \bar{A} + \bar{A}\bar{C}$

ASSORBIMENTO $\bar{A} = \bar{A} + \bar{A}\bar{B}$

COMPLEMENTARIETÀ $A + \bar{A} = 1$

IDEMPOTENZA

$$Y_7 = \bar{X}Y + \bar{X}\bar{Y}Z$$

$$= \bar{X}(Y + \bar{Y}Z)$$

$$= \bar{X}(Y + XZ + \bar{Y}Z)$$

ASSORBIMENTO $Y = Y + YZ$

$$= \bar{X}(Y + Z(\bar{X} + X))$$

COMPLEMENTARIETÀ

$$= \bar{X}(Y + Z)$$

6) ESERCIZIO 5 RISOLTO TRAMITE TABELLE DI VERITÀ - SLIDES

$$Y_1 = A + ABC + AB + BC + \bar{A}\bar{B} + \bar{A}\bar{C} \quad Y_3 = (X + Y + \bar{W})(X + Y + W)(X + \bar{Y} + W)(\bar{X} + \bar{Y} + W)$$

A	B	C	Y_1	A	B	C	Y_1
0	0	0	1	1	0	0	1
0	0	1	1	1	0	1	1
0	1	0	1	1	1	0	1
0	1	1	1	1	1	1	1

$$Y_1 = 1$$

X	Y	W	Y_3	X	Y	W	Y_3
0	0	0	0	1	0	0	1
0	0	1	0	1	0	1	1
0	1	0	0	1	1	0	0
0	1	1	1	1	1	1	1

$$Y_2 = (X + Y)(\bar{Y} + W)$$

$$Y_2 = ABC + ABC\bar{D} + \bar{A}BD + ABD + \bar{A}D$$

A	B	C	D	Y_2	A	B	C	D	Y_2	A	B	C	D	Y_2	A	B	C	D	Y_2
0	0	0	0	0	0	1	0	0	0	1	0	0	0	0	1	1	0	0	1
0	0	0	1	1	0	1	0	1	1	1	0	0	1	0	1	1	0	1	1
0	0	1	0	0	0	1	0	1	0	1	0	1	0	0	1	1	1	0	1
0	0	1	1	1	0	1	1	1	1	1	0	1	1	0	1	1	1	1	1

$\bar{A}D$

AB

$$Y_2 = \bar{A}D + AB$$

$$Y_5 = (\bar{A} + B)(A + B + D)\bar{D}$$

A	B	D	Y_5	A	B	D	Y_5
0	0	0	0	1	0	0	0
0	0	1	0	1	0	1	0
0	1	0	1	1	1	0	1
0	1	1	0	1	1	1	0

$$Y_5 = B\bar{D}$$

$$Y_7 = \bar{x}y + \bar{x}\bar{y}z$$

x	y	z	Y_7	x	y	z	Y_7
0	0	0	0	1	0	0	0
0	0	1	1	1	0	1	0
0	1	0	1	1	1	0	0
0	1	1	1	1	1	1	0

$$Y_7 = \bar{x}y + \bar{x}z$$

$$Y_6 = \bar{A}\bar{B}\bar{C}D + \bar{A}\bar{B}CD + \bar{A}B\bar{C}D + A\bar{B}\bar{C}D$$

A	B	C	D	Y_6	A	B	C	D	Y_6	A	B	C	D	Y_6	A	B	C	D	Y_6
0	0	0	0	0	0	1	0	0	0	1	0	0	0	0	1	1	0	0	0
0	0	0	1	1	0	1	0	1	1	1	0	0	1	0	1	1	0	1	0
0	0	1	0	0	0	1	1	0	0	1	0	1	0	0	1	1	1	0	0
0	0	1	1	1	0	1	1	1	0	1	0	1	1	0	1	1	1	1	0

$$Y_6 = \bar{A}\bar{B}D + \bar{A}\bar{C}D + \bar{B}\bar{C}D$$

ESERCIZIO 6 - SLIDES

A = {1: MASCHIO, 0: FEMMINA}

$\bar{A}$: DONNA

B = {1: >30 ANNI, 0: <30 ANNI}

$\bar{B}$: CON FIGLIO

C = {1: SPOSATO, 0: NON SPOSATO}

$\bar{C}$: NON SPOSATA

D = {1: HA FIGLI, 0: NON HA FIGLI}

D: <30 ANNI

F(0001) = No ASSICURATA

$$F = A\bar{B} + B\bar{D} + B\bar{D}(A+C) + B\bar{D}\bar{C}$$

A	B	C	D	F	A	B	C	D	F	A	B	C	D	F	A	B	C	D	F
0	0	0	0	0	0	1	0	0	1	1	0	0	0	1	1	1	0	0	1
0	0	0	1	0	0	1	0	1	1	1	0	0	1	1	1	1	0	1	1
0	0	1	0	0	0	1	1	0	1	1	0	1	0	1	1	1	1	0	1
0	0	1	1	0	0	1	1	1	1	1	0	1	1	1	1	1	1	1	1

RETI LOGICHE COMBINATORIE

4

ESERCIZIO 7 - SLIDES

$$Z = \overline{Z_1 Z_2}$$

$$Z_1 = Z_3 + Z_4$$

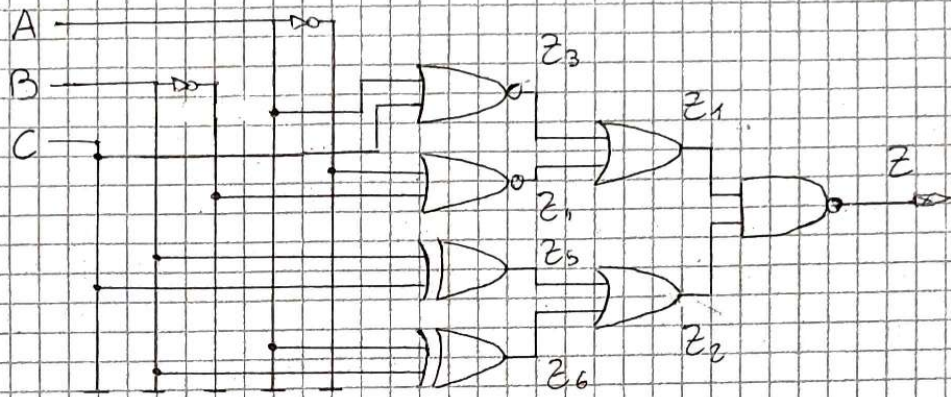
$$Z_2 = Z_5 + Z_6$$

$$Z_3 = \overline{A} \overline{C}$$

$$Z_4 = AB$$

$$Z_5 = \overline{B}C + B\overline{C}$$

$$Z_6 = \overline{A}B + A\overline{B}$$



$$Z_2 = \overline{B}C + B\overline{C} + \overline{A}B + A\overline{B}$$

$$Z_1 = \overline{A} \overline{C} + AB$$

$$Z = (\overline{B}C + B\overline{C} + \overline{A}B + A\overline{B})(\overline{A} \overline{C} + AB)$$

$$Z = (B + \overline{C})(\overline{B} + C)(A + \overline{B})(\overline{A} + B) + (A + C)(\overline{A} + B)$$

RISOLVIAMO

$$= (\overline{B}B + \overline{B}C + C\overline{B} + C\overline{C})(\overline{A}A + \overline{A}B + B\overline{A} + B\overline{B}) + (A\overline{A} + A\overline{C} + C\overline{A} + C\overline{C})$$

$$= (\overline{A}B\overline{C} + \overline{A}B\overline{C} + \overline{A}B\overline{C} + \overline{A}B\overline{C} + \overline{A}B\overline{C} + \overline{A}B\overline{C} + \overline{A}B\overline{C} + \overline{A}B\overline{C}) + (A\overline{C} + A\overline{C} + A\overline{C} + A\overline{C})$$

$$= \overline{A}B\overline{C} + \overline{A}B\overline{C} + \overline{A}B\overline{C} + \overline{A}B\overline{C} + \overline{A}B\overline{C} + \overline{A}B\overline{C} + \overline{A}B\overline{C} + \overline{A}B\overline{C}$$

$$= \overline{A}B\overline{C} + \overline{A}B\overline{C} + \overline{A}B\overline{C} + \overline{A}B\overline{C} + \overline{A}B\overline{C} + \overline{A}B\overline{C} + \overline{A}B\overline{C} + \overline{A}B\overline{C}$$

$$= \overline{A}B\overline{C} + \overline{A}B\overline{C} + \overline{A}B\overline{C} + \overline{A}B\overline{C} + \overline{A}B\overline{C} + \overline{A}B\overline{C} + \overline{A}B\overline{C} + \overline{A}B\overline{C}$$

ASSORBIMENTO $C = C + C\overline{A}$

$$= \overline{A}B\overline{C} + \overline{A}B\overline{C} + \overline{A}B\overline{C} + \overline{A}B\overline{C} + \overline{A}B\overline{C} + \overline{A}B\overline{C} + \overline{A}B\overline{C} + \overline{A}B\overline{C}$$

$$= \overline{A}B\overline{C} + \overline{A}B\overline{C} + \overline{A}B\overline{C} + \overline{A}B\overline{C} + \overline{A}B\overline{C} + \overline{A}B\overline{C} + \overline{A}B\overline{C} + \overline{A}B\overline{C}$$

$$= \overline{A}B\overline{C} + \overline{A}B\overline{C} + \overline{A}B\overline{C} + \overline{A}B\overline{C} + \overline{A}B\overline{C} + \overline{A}B\overline{C} + \overline{A}B\overline{C} + \overline{A}B\overline{C}$$

ASSORBIMENTO $\overline{B} = \overline{B} + \overline{B}AC$

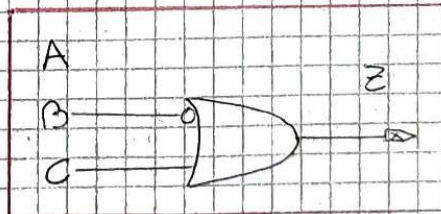
$$= \overline{B} + AC(\overline{B} + \overline{B}) + C\overline{A}$$

$$= \overline{B} + AC + C\overline{A}$$

$$= \overline{B} + C(A + \overline{A}) = \overline{B} + C$$

A	B	C	Z
0	0	0	1
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	1

→ $\overline{B} + C$



ESERCIZIO 8-SLIDES

$$Z = Z_1 + Z_2 = A + \bar{C} + \bar{B}CD$$

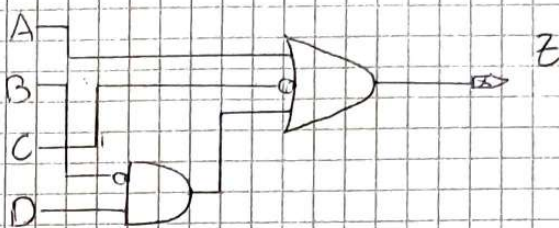
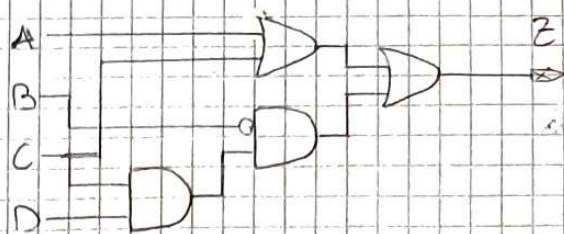
$$Z_1 = A + \bar{C}$$

$$Z_2 = \bar{B} \cdot Z_3 = \bar{B}CD$$

$$Z_3 = CD$$

Risolviamo

$$\begin{aligned} Z &= A + \bar{C} + \bar{C}\bar{B}D + \bar{B}CD \\ &= A + \bar{C} + D\bar{B}(\bar{C} + C) \\ &= A + \bar{C} + \bar{B}D \end{aligned} \quad \text{ASSORDIMENTO}$$



A	B	C	D	Z	A	B	C	D	Z	A	B	C	D	Z	A	B	C	D	Z
0	0	0	0	1	0	1	0	0	1	1	0	0	0	1	1	1	0	0	1
0	0	0	1	1	0	1	0	1	1	1	0	0	1	1	1	1	0	1	1
0	0	1	0	0	0	1	1	0	0	1	0	1	0	1	1	1	1	0	1
0	0	1	1	1	0	1	1	1	0	1	0	1	1	1	1	1	1	1	1

ESAME 4 APRILE 2023 - ES 6 PARTE 1

CALCOLA LE FORME CANONICHE DI:

$$X_1 X_2 X_0 \quad S_1 S_0 \quad X_1 X_2 X_0 \quad S_1 S_0$$

$$0 \ 0 \ 0 \ 0 \ 0 \quad 1 \ 0 \ 0 \ 0 \ 0 \ 0$$

$$0 \ 0 \ 1 \ 1 \ 0 \quad 1 \ 0 \ 1 \ 0 \ 0 \ 0$$

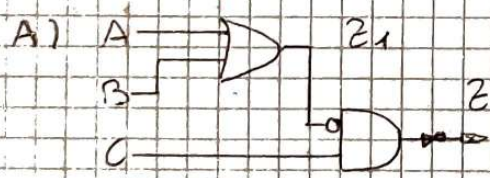
$$0 \ 1 \ 0 \ 1 \ 0 \quad 1 \ 1 \ 0 \ 0 \ 0 \ 1$$

$$0 \ 1 \ 1 \ 0 \ 1 \quad 1 \ 1 \ 1 \ 0 \ 0 \ 1$$

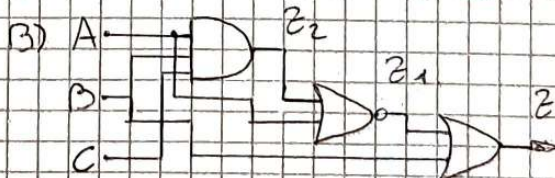
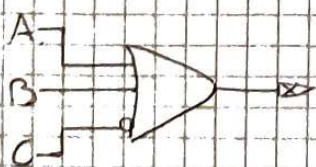
$$S_1 = \bar{X}_1 \bar{X}_2 X_0 + \bar{X}_1 X_2 \bar{X}_0$$

$$S_0 = \bar{X}_1 X_2 X_0 + X_1 X_2 \bar{X}_0 + X_1 X_2 X_0$$

ESAME 4 APRILE 2023 - ES 8 PARTE 1



$$Z = (A+B)C = A+B+C$$



$$Z = Z_1 + B = \overline{ABC} \cdot \bar{A} + B$$

$$Z_1 = \bar{Z}_2 \cdot \bar{A} = (\bar{A} + \bar{B} + \bar{C}) \bar{A} + B$$

$$Z_2 = ABC = \bar{A} + \bar{B}A + \bar{C}A + B$$

$$= \bar{A}(1 + \bar{B} + \bar{C}) + B = \bar{A} + B$$



ESAME 4 APRILE 2023 - ES 16 PARTE 2

$$X_2 X_1 X_0 \quad S_1 S_0 \quad X_2 X_1 X_0 \quad S_1 S_0$$

$$0 \ 0 \ 0 \ 0 \ 1 \quad 1 \ 0 \ 0 \ 1 \ 1$$

$$0 \ 0 \ 1 \ 1 \ 0 \quad 1 \ 0 \ 1 \ 0 \ 0$$

$$0 \ 1 \ 0 \ 1 \ 0 \quad 1 \ 1 \ 0 \ 1 \ 1$$

$$0 \ 1 \ 1 \ 0 \ 1 \quad 1 \ 1 \ 1 \ 0 \ 1$$

$$S_1 = \bar{X}_2 \bar{X}_1 X_0 + \bar{X}_2 X_1 \bar{X}_0 + X_2 \bar{X}_1 X_0 + X_2 X_1 \bar{X}_0$$

$$S_2 = \bar{X}_2 \bar{X}_1 X_0 + \bar{X}_2 X_1 X_0 + X_2 \bar{X}_1 X_0 + X_2 X_1 \bar{X}_0 + X_2 X_1 X_0$$

RETI LOGICHE COMBINATORIE

5

ESAME 8 APRILE ⁽²⁰²⁴⁾ ESSE E PARTE I

1) $F = \bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}\bar{B}C\bar{D} + \bar{A}B\bar{C}\bar{D} + A\bar{B}C\bar{D}$

A) SÌ, È IN FORMA CANONICA.

B) ECCO LA TABELLA DI VERITÀ:

$$F = \bar{A}\bar{B}\bar{D}(\bar{C} + C) + \bar{A}\bar{D}(\bar{B}\bar{C} + B\bar{C})$$

$$= \bar{A}\bar{B}\bar{D} + \bar{A}\bar{B}C\bar{D} + \bar{A}B\bar{C}\bar{D}$$

$$= \bar{B}\bar{D}(\bar{A} + A\bar{C}) + \bar{A}B\bar{C}\bar{D}$$

$$= \bar{B}\bar{D}(\bar{A} + \bar{A}\bar{C} + A\bar{C}) + \bar{A}B\bar{C}\bar{D}$$

ASSORBIMENTO $\bar{A} = \bar{A} + \bar{A}\bar{C}$

$$= \bar{B}\bar{D}(\bar{A} + \bar{C}(A + \bar{A})) + \bar{A}B\bar{C}\bar{D}$$

$$= \bar{B}\bar{D}\bar{A} + \bar{B}\bar{D}\bar{C} + \bar{A}B\bar{C}\bar{D}$$

$$0 \ 1 \ 0 \ 0 \ 0$$

$$0 \ 1 \ 0 \ 1 \ 0$$

$$0 \ 1 \ 1 \ 0 \ 0$$

$$0 \ 1 \ 1 \ 1 \ 0$$

$$1 \ 0 \ 0 \ 0 \ 1$$

$$1 \ 0 \ 0 \ 1 \ 0$$

$$1 \ 0 \ 1 \ 0 \ 0$$

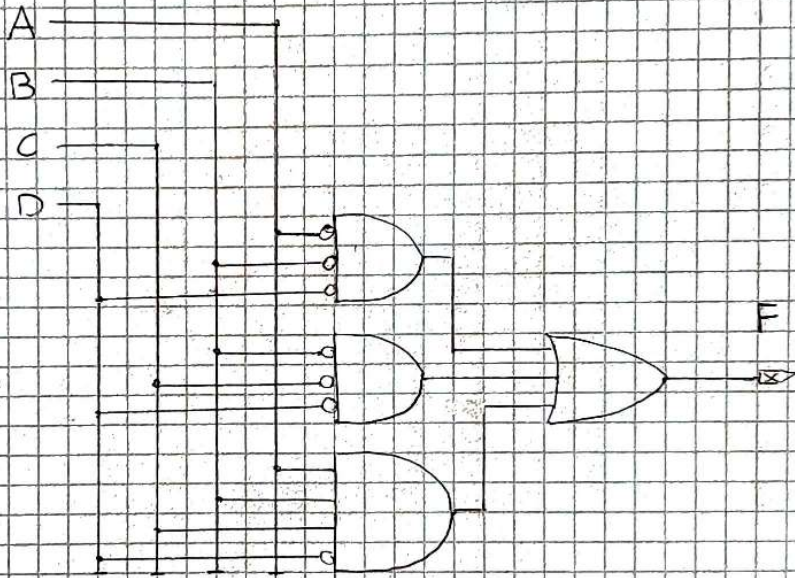
$$1 \ 0 \ 1 \ 1 \ 0$$

$$1 \ 1 \ 0 \ 0 \ 0$$

$$1 \ 1 \ 0 \ 1 \ 0$$

$$1 \ 1 \ 1 \ 0 \ 1$$

$$1 \ 1 \ 1 \ 1 \ 0$$



ESAME 8 APRILE 2024 - ES 6 E 8 PARTE 2

$$F = \bar{A}\bar{B}\bar{C}D + \bar{A}B\bar{C}\bar{D} + A\bar{B}\bar{C}D + A\bar{B}C\bar{D} + ABCD$$

A) SÌ, È IN FORMA CANONICA, I MINTERMINI HANNO TUTTI 4 VARIABILI.

B) ECCO LA TABELLA DI VERITÀ:

A	B	C	D	F	
0	0	0	0	0	$= \bar{B}\bar{C}D(\bar{A} + A) + \bar{A}B\bar{C}\bar{D} + A\bar{B}\bar{C}D + A\bar{B}C\bar{D} + ABCD$
0	0	0	1	1	$= \bar{B}\bar{C}D + \bar{A}B\bar{C}\bar{D} + A\bar{B}\bar{C}D + A\bar{B}C\bar{D} + ABCD$
0	0	1	0	1	
0	0	1	1	0	
0	1	0	0	0	
0	1	0	1	0	
0	1	1	0	1	
0	1	1	1	0	
1	0	0	0	0	
1	0	0	1	0	
1	0	1	0	0	
1	0	1	1	0	
1	1	0	0	1	
1	1	0	1	0	
1	1	1	0	0	
1	1	1	1	1	

